

Math 526, F09
Quiz 5

Name: Answer
SSN: xxx-xx-

Instructions: There is 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Let $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix}$.

(a) Reduce A to its ordinary echelon form U .

$$A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ -2 & -6 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix}$$
$$\rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

(b) Which are the free variables and which are the pivot variables?

pivot variables: x_1, x_3

free variables: x_2, x_4

(c) Find the null space of A , $N(A)$.

$$UX = 0 \Rightarrow \begin{cases} x_1 + 3x_2 + x_3 + 2x_4 = 0 \\ 2x_3 + 4x_4 = 0 \end{cases}$$

$$\text{Set } x_2 = 1, x_4 = 0, \text{ we get } \begin{cases} x_1 + x_3 = -3 \\ 2x_3 = 0 \end{cases} \Rightarrow \begin{matrix} x_1 = -3 \\ x_3 = 0 \end{matrix}$$

$$\Rightarrow s_1 = (-3, 1, 0, 0)$$

$$\text{Set } x_2 = 0, x_4 = 1, \text{ we get } \begin{cases} x_1 + x_3 = -2 \\ 2x_3 = -4 \end{cases} \Rightarrow \begin{matrix} x_1 = 0 \\ x_3 = -2 \end{matrix}$$

$$\Rightarrow s_2 = (0, 0, -2, 1) \Rightarrow x = c_1 s_1 + c_2 s_2 = \begin{bmatrix} -3c_1 \\ c_1 \\ -2c_2 \\ c_2 \end{bmatrix}$$

$$\text{Thus } N(A) = \left\{ \begin{bmatrix} -3c_1 \\ c_1 \\ -2c_2 \\ c_2 \end{bmatrix} \mid c_1, c_2 \text{ are real numbers} \right\}$$